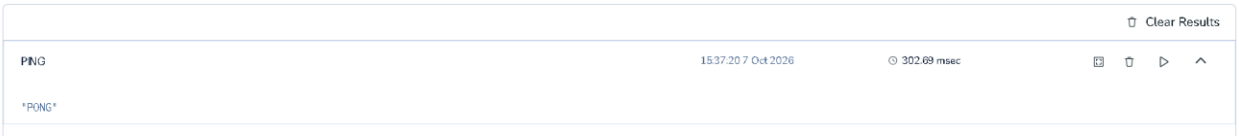
Bài làm

1. Lab 1: Quản Lý Caching Dữ Liệu & Phiên Đăng Nhập



1: Kết quả ping



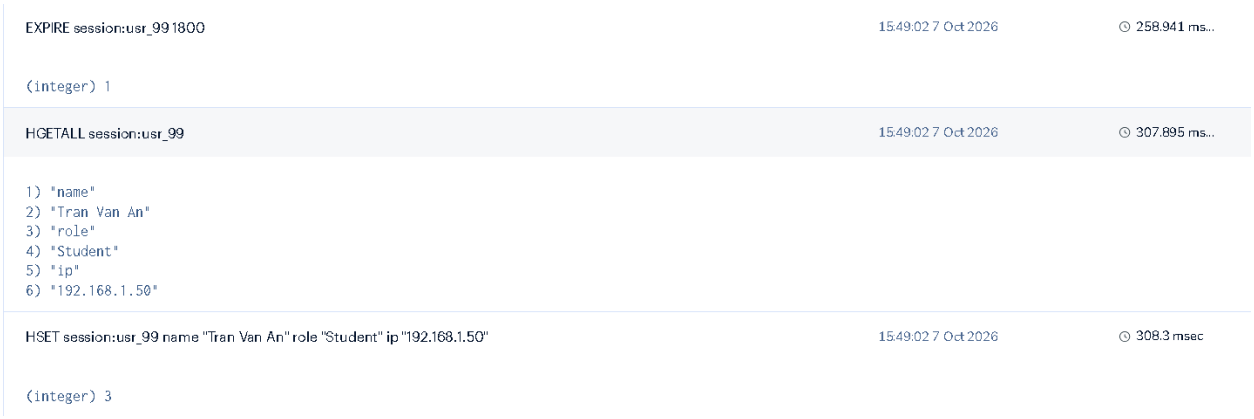
2: Xem số lượng Key đang lưu trong DB



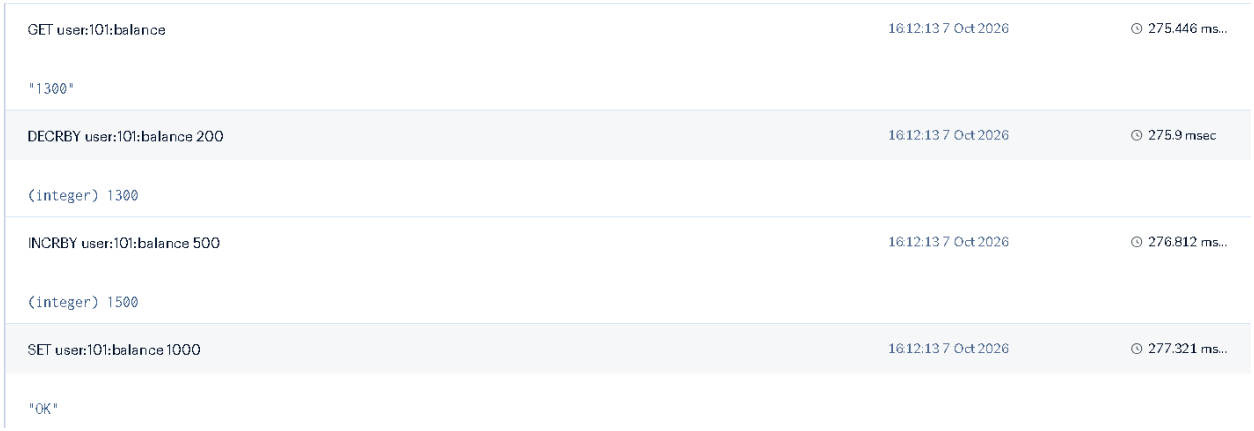
3: Quản lý lượt xem trang web



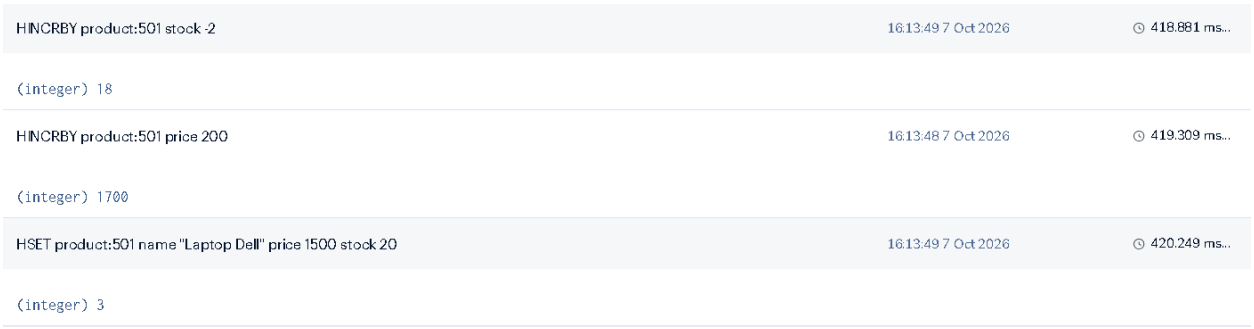
4: Lưu mã xác thực OTP và Lấy nhiều giá trị thuộc tính cùng lúc



5: Lưu thông tin phiên làm việc



6: Tạo, Nạp, Rút, Hiển thị số dư



7: Lưu thông tin sản phẩm, Tăng giá, Giảm tồn kho

```
HGETALL product:501                                                    16:13:49 7 Oct 2026    418.516 ms...

1) "name"
2) "Laptop Dell"
3) "price"
4) "1700"
5) "stock"
6) "18"
```

8: Hiển thị thông tin giá và tồn kho mới

2. Lab 2 : Hàng đợi (Queue) & Bảng xếp hạng (ZSET)

```
ZADD student:scores 8.5 "Nguyen Van A" 9.0 "Tran Thi B" 7.5 "Le Van C" 9.8 "Pham Minh D"    16:23:01 7 Oct 2026    281.178 ms...

(integer) 4

RPOP queue:email_tasks                                                 16:23:01 7 Oct 2026    282.003 ms...

"email_usr1"

LLEN queue:email_tasks                                                 16:23:01 7 Oct 2026    282.378 ms...

(integer) 3

LPUSH queue:email_tasks "email_usr1" "email_usr2" "email_usr3"         16:23:01 7 Oct 2026    282.885 ms...

(integer) 3
```

9: Cơ chế hàng đợi và Xếp hạng sinh viên môn NoSql

```
ZINCRBY student:scores 1.0 "Le Van C"                                  16:23:02 7 Oct 2026    262.179 ms...

"8.5"

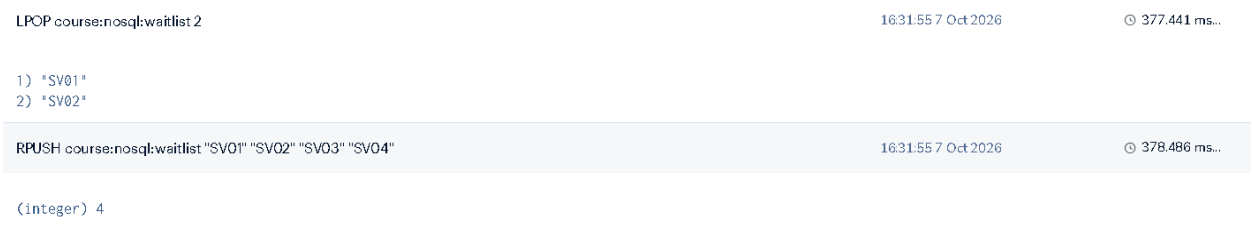
ZREVRANGE student:scores 0 -1 WITHSCORES                               16:23:01 7 Oct 2026    280.633 ms...

1) "Pham Minh D"
2) "9.8"
3) "Tran Thi B"
4) "9"
5) "Nguyen Van A"
6) "8.5"
7) "Le Van C"
8) "7.5"
```

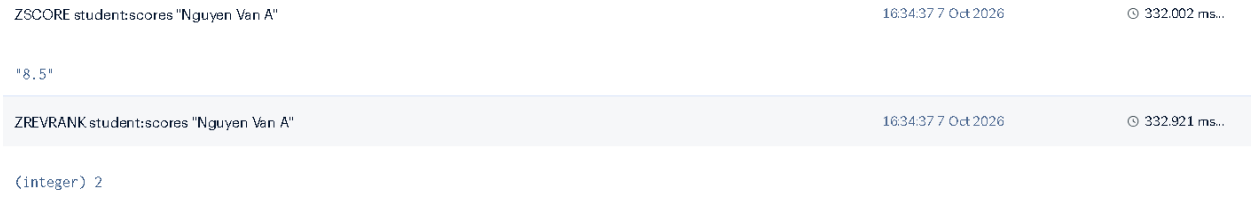
10: Lấy toàn bộ xếp hạng từ điểm cao xuống thấp kèm điểm số và Cộng 1.0 điểm cho sinh viên Le Van C



11: Lấy danh sách sinh viên từ 8.0 đến 10.0 điểm

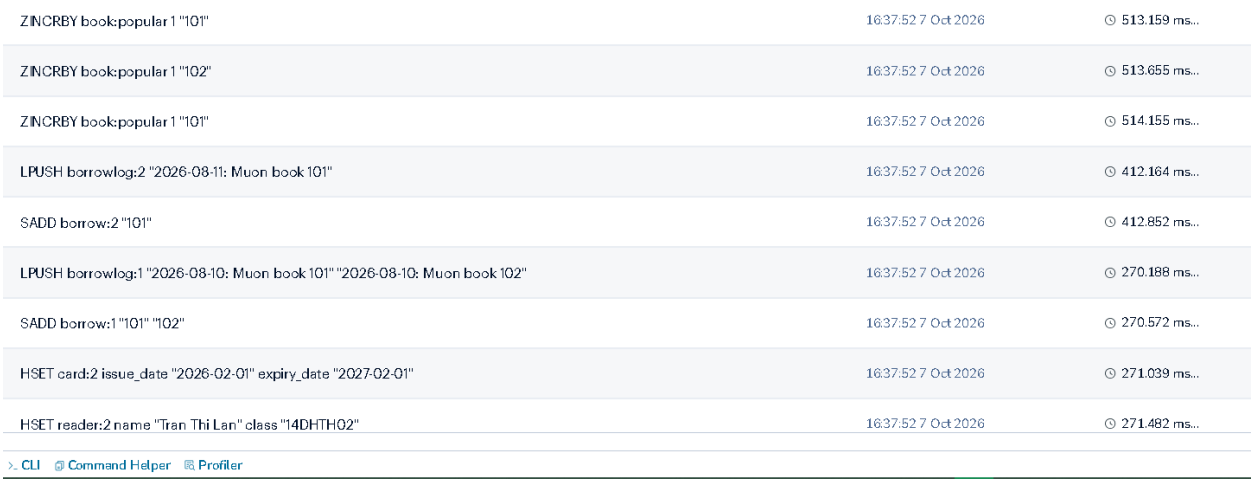


12: Thêm sinh viên vào danh sách chờ và Lấy 2 sinh viên đăng ký sớm nhất



13: Lấy vị trí thứ hạng giảm dần và Xem điểm số trực tiếp của sinh viên đó

3. Lab 3: Bài toán tổng hợp quản lý thư viên HUIT



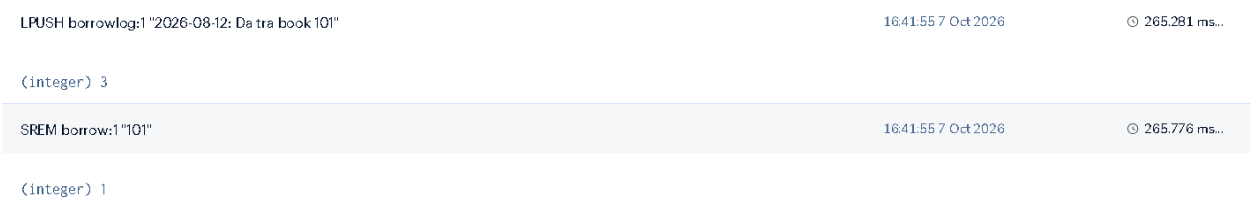
14: Khởi tạo dữ liệu mẫu cho thư viện



15: Lấy tất cả các phần tử trong Set sách mượn của reader:1

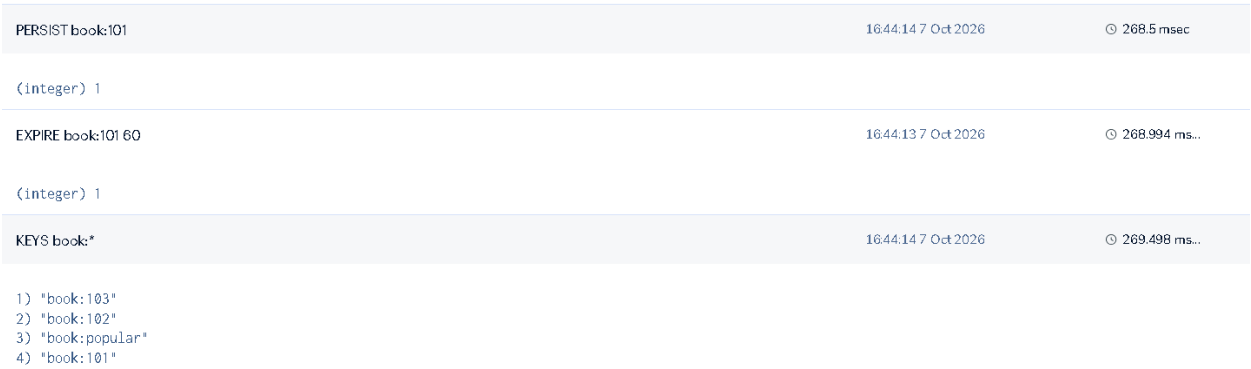


16: Lấy TOP 3 sách mượn nhiều nhất kèm số lượt mượn

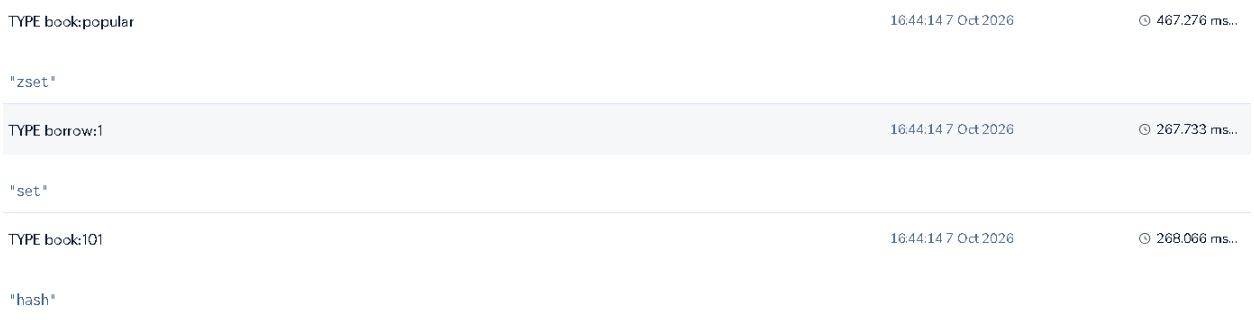


17: Viết câu lệnh xử lý khi Độc giả 1 trả cuốn sách `101`

4. Quản lý thời gian tồn tại & Pattern Search

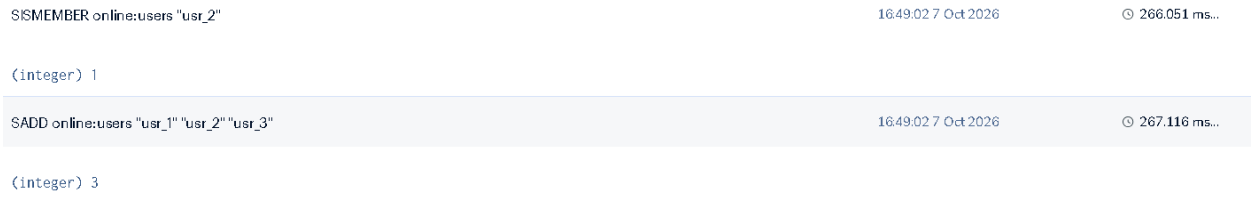


18: Tìm kiếm key dạng 'book:', Đặt key sống trong 60 giây, Loại bỏ thời gian hết hạn giữ key sống vĩnh viễn

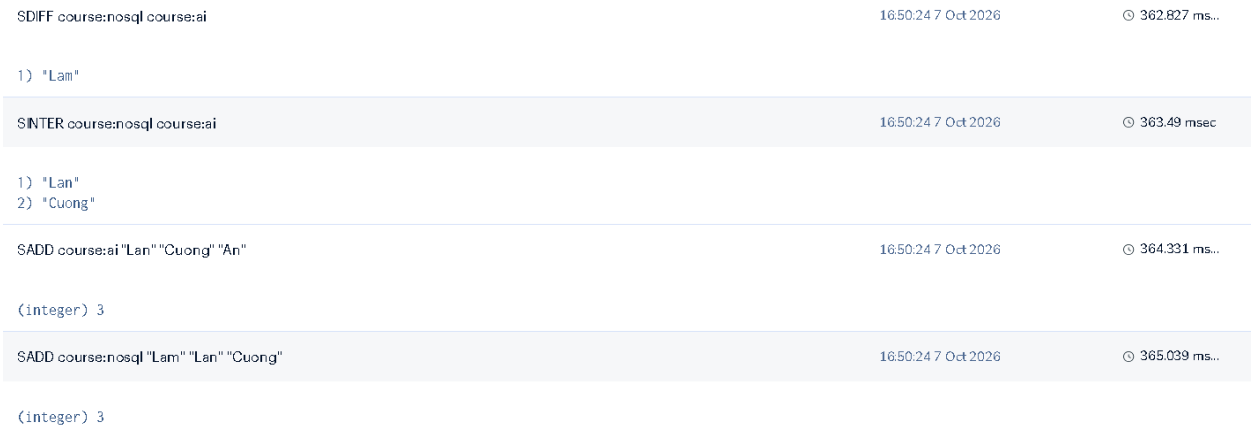


19: Kiểm tra kiểu dữ liệu của key

5. Bài tập tự luyện tại lớp

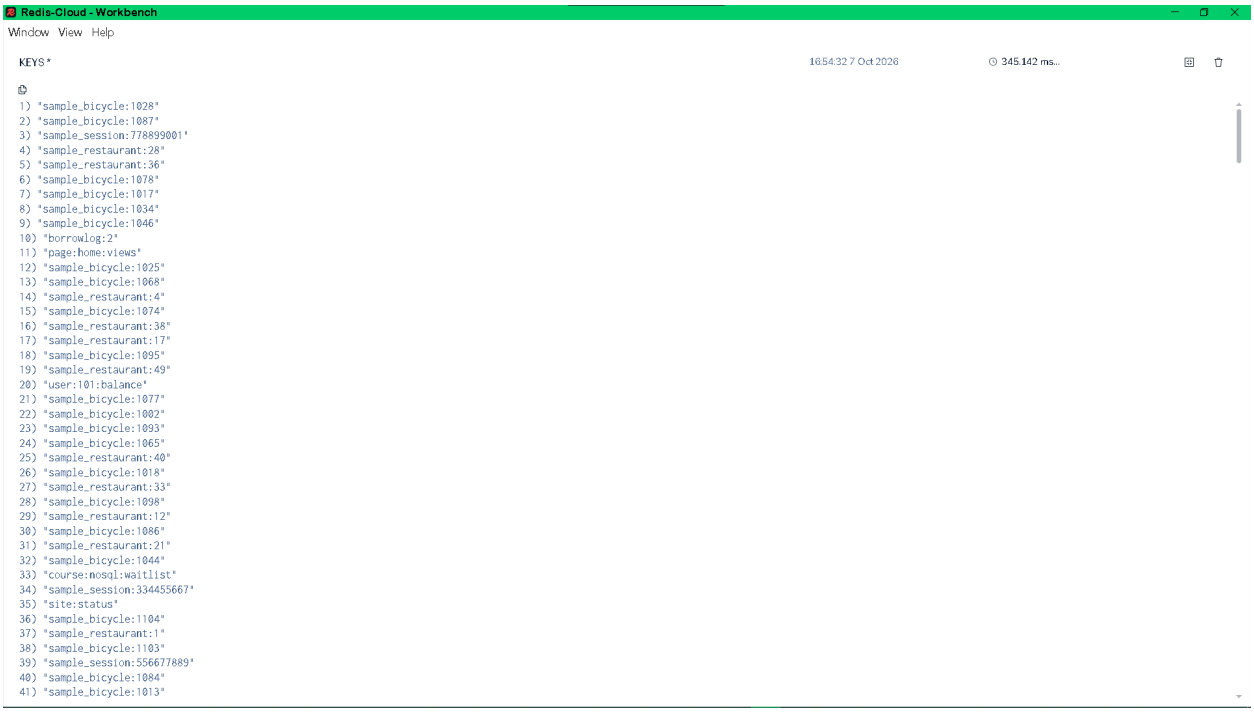


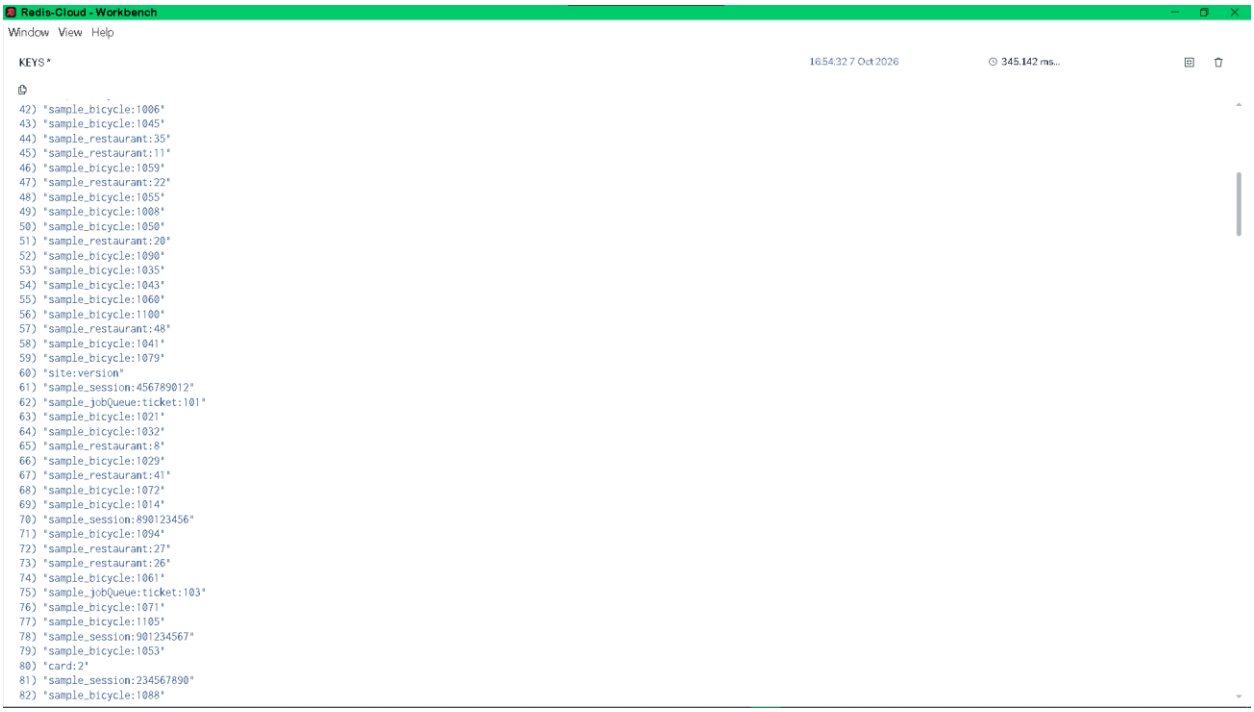
20: Bài 4.1

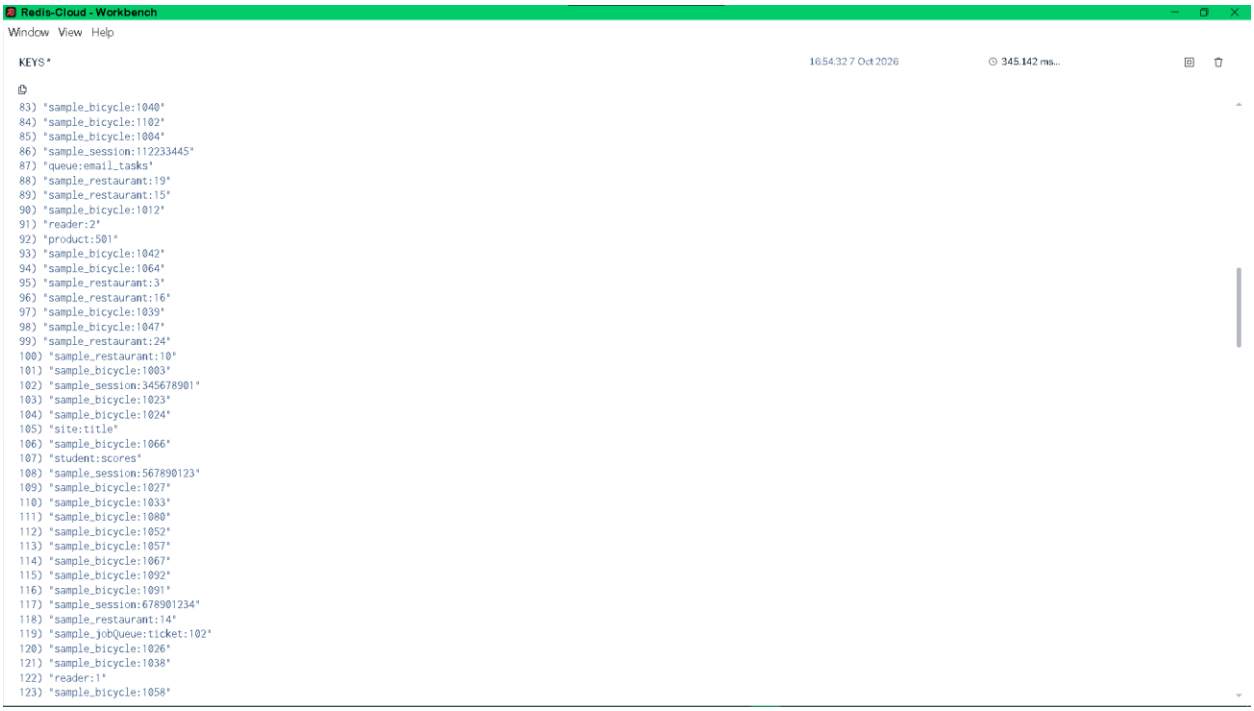


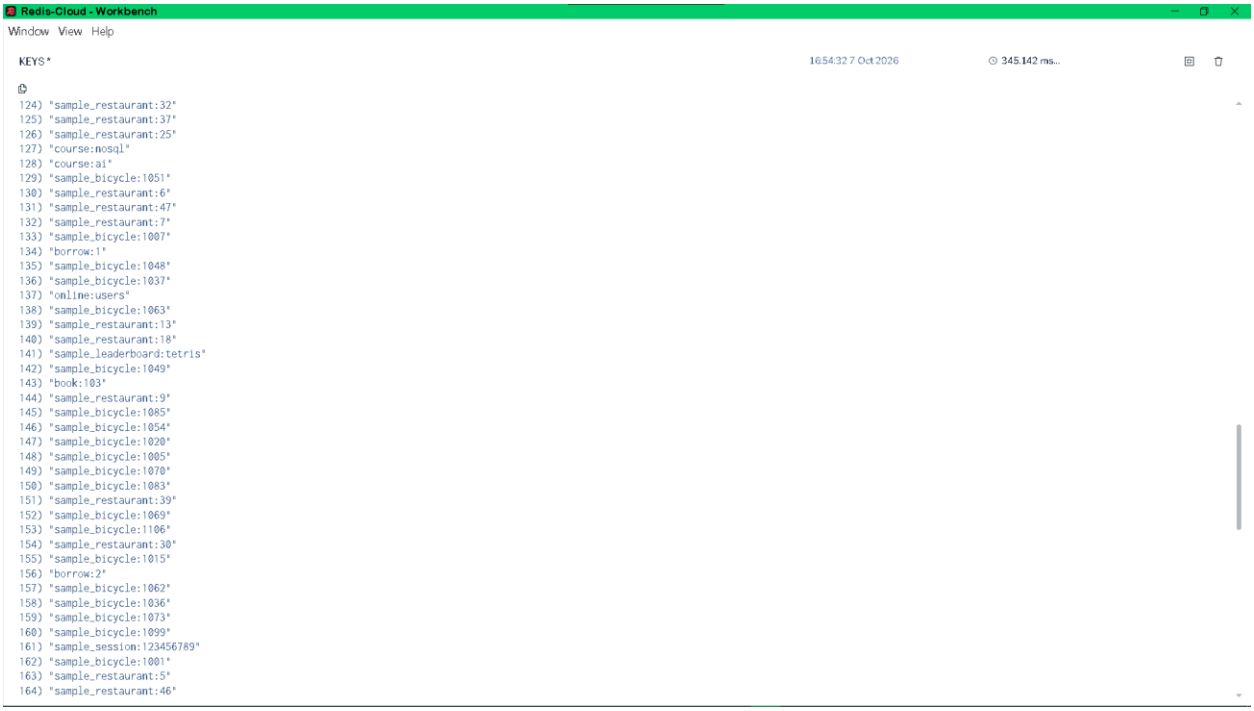
21: Bài 4.2

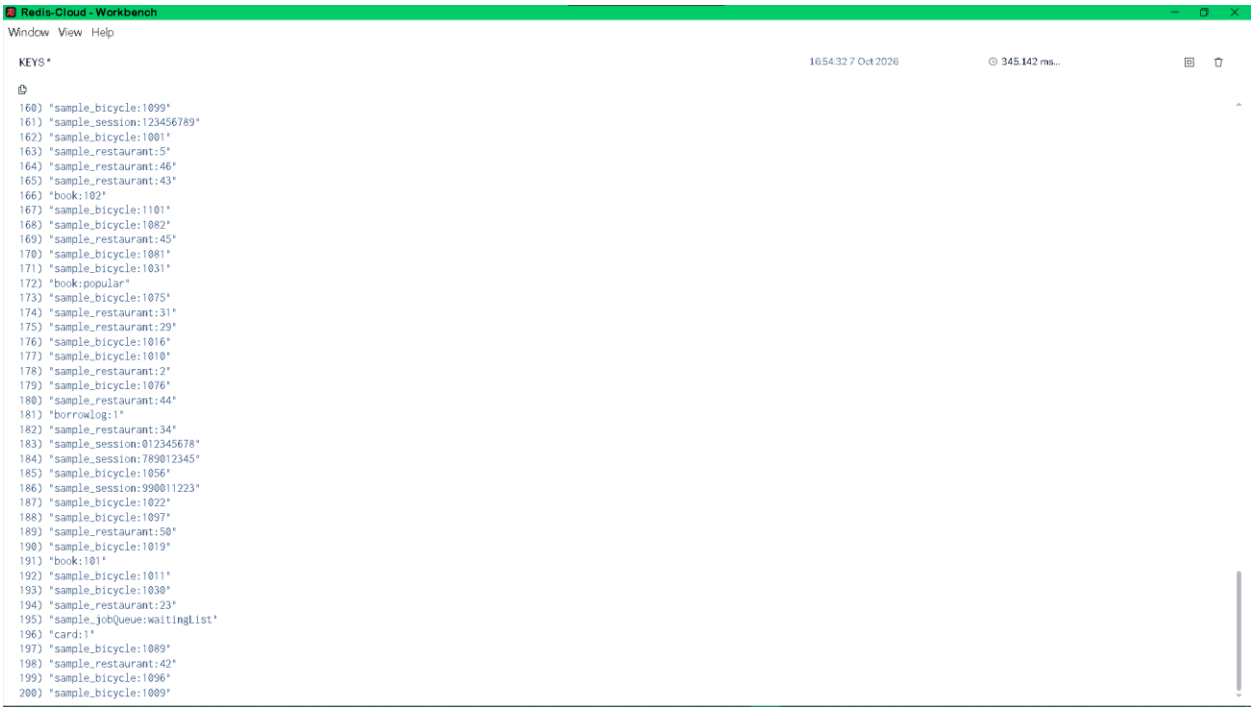
6. Kết quả chạy lệnh KEYS*:






Redis-Cloud - Workbench
Window View Help
KEYS *
16:54:32 7 Oct 2026
345.142 ms...
83) "sample_bicycle:1040"
84) "sample_bicycle:1102"
85) "sample_bicycle:1004"
86) "sample_session:112233445"
87) "queue:email_tasks"
88) "sample_restaurant:19"
89) "sample_restaurant:15"
90) "sample_bicycle:1012"
91) "reader:2"
92) "product:501"
93) "sample_bicycle:1042"
94) "sample_bicycle:1064"
95) "sample_restaurant:3"
96) "sample_restaurant:16"
97) "sample_bicycle:1039"
98) "sample_bicycle:1047"
99) "sample_restaurant:24"
100) "sample_restaurant:10"
101) "sample_bicycle:1003"
102) "sample_session:345678901"
103) "sample_bicycle:1023"
104) "sample_bicycle:1024"
105) "site:title"
106) "sample_bicycle:1066"
107) "student:scores"
108) "sample_session:567890123"
109) "sample_bicycle:1027"
110) "sample_bicycle:1033"
111) "sample_bicycle:1080"
112) "sample_bicycle:1052"
113) "sample_bicycle:1057"
114) "sample_bicycle:1067"
115) "sample_bicycle:1092"
116) "sample_bicycle:1091"
117) "sample_session:678901234"
118) "sample_restaurant:14"
119) "sample_jobQueue:ticket:102"
120) "sample_bicycle:1026"
121) "sample_bicycle:1038"
122) "reader:1"
123) "sample_bicycle:1058"


Redis-Cloud - Workbench
Window View Help
KEYS *
16:54:32 7 Oct 2026
345.142 ms...
124) "sample_restaurant:32"
125) "sample_restaurant:37"
126) "sample_restaurant:25"
127) "course:nosql"
128) "course:ai"
129) "sample_bicycle:1051"
130) "sample_restaurant:6"
131) "sample_restaurant:47"
132) "sample_restaurant:7"
133) "sample_bicycle:1007"
134) "borrow:1"
135) "sample_bicycle:1048"
136) "sample_bicycle:1037"
137) "online:users"
138) "sample_bicycle:1063"
139) "sample_restaurant:13"
140) "sample_restaurant:18"
141) "sample_leaderboard:tetris"
142) "sample_bicycle:1049"
143) "book:103"
144) "sample_restaurant:9"
145) "sample_bicycle:1085"
146) "sample_bicycle:1054"
147) "sample_bicycle:1020"
148) "sample_bicycle:1005"
149) "sample_bicycle:1070"
150) "sample_bicycle:1083"
151) "sample_restaurant:39"
152) "sample_bicycle:1069"
153) "sample_bicycle:1106"
154) "sample_restaurant:30"
155) "sample_bicycle:1015"
156) "borrow:2"
157) "sample_bicycle:1062"
158) "sample_bicycle:1036"
159) "sample_bicycle:1073"
160) "sample_bicycle:1099"
161) "sample_session:123456789"
162) "sample_bicycle:1001"
163) "sample_restaurant:5"
164) "sample_restaurant:46"


Redis-Cloud - Workbench
Window View Help
KEYS *
16:54:32 7 Oct 2026
345.142 ms...
160) "sample_bicycle:1099"
161) "sample_session:123456789"
162) "sample_bicycle:1001"
163) "sample_restaurant:5"
164) "sample_restaurant:46"
165) "sample_restaurant:43"
166) "book:102"
167) "sample_bicycle:1101"
168) "sample_bicycle:1082"
169) "sample_restaurant:45"
170) "sample_bicycle:1081"
171) "sample_bicycle:1031"
172) "book:popular"
173) "sample_bicycle:1075"
174) "sample_restaurant:31"
175) "sample_restaurant:29"
176) "sample_bicycle:1016"
177) "sample_bicycle:1010"
178) "sample_restaurant:2"
179) "sample_bicycle:1076"
180) "sample_restaurant:44"
181) "borrowlog:1"
182) "sample_restaurant:34"
183) "sample_session:012345678"
184) "sample_session:789012345"
185) "sample_bicycle:1056"
186) "sample_session:990011223"
187) "sample_bicycle:1022"
188) "sample_bicycle:1097"
189) "sample_restaurant:50"
190) "sample_bicycle:1019"
191) "book:101"
192) "sample_bicycle:1011"
193) "sample_bicycle:1030"
194) "sample_restaurant:23"
195) "sample_jobQueue:waitingList"
196) "card:1"
197) "sample_bicycle:1089"
198) "sample_restaurant:42"
199) "sample_bicycle:1096"
200) "sample_bicycle:1009"